Introduction to Calculations of Heat loss on piping

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1 Introduction

This excel sheet allows to calculate heat losses for a single pipe with flowing liquid. It utilizes a Python module available in Excel environment. The exact usage of this packet is described in the sheet "User's Guide". There are two modes of calculation:

- Normal - it is used for any calculations that do not involve any additional external heat flux.
- Heat flux - it is used only if the piping is exposed to constant heat flux.

Those methods are described in more details in the next section.

2 Theory behind calculations

Both modes are utilizing the same energy balance and methodology. There are special additional considerations when it comes down to solar radiation mode, which are later explained. Here are main assumptions that allows us to solve mass and energy balance:

1. Steady-state process.
2. The change in kinetic and potential energy is negligible.
3. The flow is turbulent and pipes are full of liquid.
4. The fluid is incompressible.
5. The specific heat capacity is constant.
6. Fluid is loosing energy to surroundings.
7. No axial heat transfer.
8. Uniform heat transfer distribution in piping cross section.
9. Piping is horizontal and enclosed by surface at ambient temperature separated by air only at ambient pressure and temperature.
10. Internal fluid heat transfer coeff. is infinite and material conductivities are constant.
11. Heat flux is from outside of piping with absorptivity of 0.25 (heat flux mode only).

With above in mind, one can utilize "Thermal Resistance Network" methodology and write that the average heat loss rate through a pipe segment of length L and distance x from the inlet is:

$$\dot{Q}_{avg}(x) = \frac{(T_f(x) - T_a)_{avg}}{R_{fa}} = 2\pi LK \left(T_f(x) - T_a\right)_{avg} \tag{1}$$

The *avg* notation means that we should take some weighted average of the temperature difference on piping segment of length L. With the same assumptions we can write that energy balance at any point must satisfy:

$$A_t \dot{q}_o - \dot{Q} = \dot{m} C dT_f \tag{2}$$

where:

- $T_f(x)$ - temperature of liquid at distance x, (°F, °C)
- T_a - ambient temperature, (°F, °C)
- A_t - total piping area (ft^2, m^2)
- $\dot{q}_o$ - incident heat flux $\left(\frac{\text{BTU}}{\text{ft}^2\text{h}}, \frac{\text{W}}{\text{m}^2}\right)$
- R_{fa} - thermal resistance of piping of length from fluid to air L, $\left(\frac{^\circ\text{F}}{\text{BTU}}, \frac{^\circ\text{C}}{\text{W}}\right)$
- K - combined thermal conductivity, $\left(\frac{\text{BTU}}{\text{ft}\cdot\text{h}\cdot^\circ\text{F}}, \frac{\text{W}}{\text{m}\cdot\text{K}}\right)$
- $\dot{m}$ - mass flow rate, $\left(\frac{\text{lb}}{\text{h}}, \frac{\text{kg}}{\text{s}}\right)$
- C - specific heat capacity, $\left(\frac{\text{BTU}}{\text{lb}\cdot\text{R}}, \frac{\text{kJ}}{\text{kg}\cdot\text{K}}\right)$

and we define K as:

$$\frac{1}{K} = \frac{1}{k_{pipe}} ln\left(\frac{D_o}{D_i}\right) + \frac{1}{k_{ins}} ln\left(\frac{D_o + 2y}{D_o}\right) + \frac{2N}{h(D_o + 2y)} \tag{3}$$

where:

- N - length conversion $\left(eq.\ 12\frac{\text{inch}}{\text{ft}}\right)$
- y - insulation thickness, inch
- D_i - inner diameter, inch
- D_o - outer diameter, inch
- h - combined heat transfer coefficient from surface of a piping to air, $\left(\frac{\text{BTU}}{\text{ft}^2\cdot\text{h}\cdot^\circ\text{F}}, \frac{\text{W}}{\text{m}^2\cdot\text{K}}\right)$
- k_{pipe} - thermal conductivity of a piping, $\left(\frac{\text{BTU}}{\text{ft}\cdot\text{h}\cdot^\circ\text{F}}, \frac{\text{W}}{\text{m}\cdot\text{K}}\right)$
- k_{ins} - thermal conductivity of a insulation, $\left(\frac{\text{BTU}}{\text{ft}\cdot\text{h}\cdot^\circ\text{F}}, \frac{\text{W}}{\text{m}\cdot\text{K}}\right)$

2.1 General solution

The presented approach does not calculate fluid temperature readily. Instead, surface temperature at $x = 0$ length is found, then integration is carried out to find surface temperature profile with length, and finally it is translated back to fluid temperature.

Before presenting general solution, it is important to remember that K and h are inherently functions of surface temperature T_s, which will be stated explicitly from now on. Let us also define three properties that simplifies the solution form.

$$\lambda = \frac{\dot{m}C}{2\pi}, \quad \text{F}(T_s) = \frac{D_t h(T_s)}{2NK(T_s)}, \quad \text{B}(T_s) = \frac{2\dot{q}_o}{h(T_s)\Delta T_{sa}}$$

With properties above, it can be shown that, there exist a set of parametric curves satisfying both equations (1) and (2) of form:

$$\text{F}(T_s)\left(\Delta T_{sa} - \frac{\dot{q}_o}{h(T_s)}\right) - \Delta T_{fa} = 0 \tag{4}$$

$$\int_{T_{s,0}}^{T_s} \frac{1}{K(T_s)\,(1-\mathrm{B}(T_s))} \left[\left(1 + \dot{q}_o \frac{d}{dT_s}\frac{1}{h(T_s)}\right) \frac{1}{\Delta T_{sa}} + \left(1 - \frac{\mathrm{B}(T_s)}{2}\right) \frac{d\ln \mathrm{F}(T_s)}{dT_s} \right] dT_s + \frac{x}{\lambda} = 0 \tag{5}$$

With $\Delta T_{ia} = (T_i - T_a)$. Because above integral does not have a closed form solution, it must be solved with different methods.

The curve (5) is solved with minimization algorithms. Before stating it mathematically, let us rewrite integrand as $\mathrm{I}(T_s)$, then we can construct so-called loss function of form:

$$\mathcal{L}(x, T_s) = \left(\int_{T_{s,0}}^{T_s} \mathrm{I}(T_s) dT_s + \frac{x}{\lambda} \right)^2$$

And the final solution is found by minimizing this function on specially constructed interval (denoted Y), which is breafly explained in chapter 2.4. Mathematically, the surface temperature at given length is found by solving:

$$T_s(x) = \underset{T \in \mathrm{Y}}{\operatorname{argmin}}\ \mathcal{L}(x, T) \tag{6}$$

This value is then put back to (4) to recover fluid temperature.

2.2 Heat Transfer coefficient of piping surface

Heat transfer coefficient of air film around piping surface is calculated as a sum of radiation and convection effects. This provides a straightforward way of calculating both effects separately. Radiation effect is calculated with Stefan-Boltzman law, whereas convection effect is calculated with Churchill and Chu correlation for horizontal piping. That is:

$$\begin{aligned} h(T_s) &= h_{rad}(T_s) + h_{conv}(T_s) \\ h_{rad}(T_s) &= \epsilon \cdot 1.71 \cdot 10^{-9} \cdot (T_s^2 + T_a^2) \cdot (T_s + T_a) \\ h_{conv}(T_s) &= 9,636.27 \cdot \frac{\mu(T_{fl}) \cdot C_p(T_{fl})}{D_t} \left(0.6 + 0.321 \cdot \mathrm{Ra}^{1/6}\right)^2 \end{aligned} \tag{7}$$

where the equation for convective heat transfer coefficient assumes $\mathrm{Pr} = 0.72$ for all range of temperatures. Here, T_{fl}, μ and C_p are film temperature, dynamic viscosity and specific heat capacity of air at insulation-air interface, respectively, and D_t is total diameter in inch of piping segment. All of below equations are defined for SI units only. Users can use both Imperial and SI for calculations, and internally the python module uses SI metric for equations (8), (9) and (10).

Film temperature is just a mean value between surface temperature and ambient temperature:

$$T_{fl} = \frac{T_s + T_a}{2}\,[\mathrm{K}] \tag{8}$$

We calculate dynamic viscosity from Sutherland's law for air:

$$\mu(T) = \frac{1.458 \cdot 10^{-6} \cdot T^{1.5}}{T + 110.4} [\mathrm{Pa \cdot s}] \tag{9}$$

Polynomial relation for specific heat of air is taken from Cengel, Y. A., Boles, M. A., & Kanoglu, M. (2015). "Thermodynamics: An engineering approach" (8th ed.):

$$C_p(T) = 0.9703 + 6.7898 \cdot 10^{-5} \cdot T + 1.6576 \cdot 10^{-7} \cdot T^2 - 6.7863 \cdot 10^{-11} \cdot T^3 \left[\frac{\mathrm{kJ}}{\mathrm{kg \cdot K}}\right] \tag{10}$$

Above equations (9) and (10) are valid in a range from around 200K to 1800K. There is no limit on upper or lower temperature in the spreadsheet, but below 273K specific heat of air is assumed to be equal to $C_p(273K)$ and above 1800K to be equal to $C_p(1800K)$.

In equation (7) Ra is the Rayleigh number which for air at low reduced pressure (below 0.1) can be calculated from:

$$\mathrm{Ra} = \frac{g\beta(T_s - T_a)D_t^3}{\nu\alpha} = 14.43 \cdot \frac{T_s - T_a}{T_{fl}^3 \cdot \mu(T_{fl})^2} D_t^3 \tag{11}$$

Here, g, β and α is the gravitational acceleration, isobaric thermal expansion coefficient and thermal diffusitivty of air, repspectively. This exact equation is obtained by assuming ambient pressure of 1 atm.

2.3 Heat Transfer Rate Calculation

There are three different values of heat transfer rate per segment length reported back to user. Those are:

- Heat Transfer Total ($\dot{q}_{loss}$) - This is total heat transfer that is being lost by the fluid. This means that if fluid temperature decreases this value is positive and if fluid temperature is increasing this value is negative. It is realated to fluid properties by $x_t \dot{q}_{loss} = -\dot{m} C \Delta T_f$.
- Heat Transfer by Radiation/Convection ($\dot{q}_{rad/conv}$) - This is total heat transfer from the piping surface. Those values are positive if inlet fluid temperature is greater than ambient temperature, and negative if fluid temperature is less than ambient temperature.

While total heat transfer is easily related to known values, this is not the case for heat transfer by radiation and convection. The form of (5) requires us to utilize definition equation (1) at some weighted average surface temperature, instead of typical LMTD approach. Given energy balance of piping surface it can be shown that:

$$\dot{\mathbf{q}} = \frac{\pi D_t}{N} \mathbf{h}(T_{s,avg}) \Delta T_{sa,avg} \tag{12}$$

Where $\dot{\mathbf{q}} = (\dot{q}_{rad}, \dot{q}_{conv})$ and $\mathbf{h}(T_{s,avg}) = (h_{rad}(T_{s,avg}), h_{conv}(T_{s,avg}))$ and solution is obtained by the use of minimization algorithms similar to (6). It is important to remember that sum of radiative and convective heat transfer rates will be equal to total heat transfer only and only if heat flux is zero.

2.4 Minimization interval Y

Under assumptions mentioned in chapter 2, there are three cases that can be considered when solving (5):

1. $\dot{q} = 0$
2. $0 < \dot{q}_o < 0.5 \cdot h(T_{s,0}) \Delta T_{sa,0}$
3. $\dot{q}_o > 0.5 \cdot h(T_{s,0}) \Delta T_{sa,0}$

Here, (1) is easy to understand and Y is simply the bounds one would naturally choose:

$$\mathrm{Y} = [T_a, T_{s,0}] \tag{13}$$

In case of (2), the surface temperature is still decreasing with increasing length, but the lower bound is no longer ambient temperature due to incident radiation:

$$\mathrm{Y} = [T, T_{s,0}], \text{ where } T \text{ s.t. } h(T)(T - T_a) = 2\dot{q}_o \tag{14}$$

In the last case (3), surface temperature increases with increasing length - fluid is heated by external heat flux. The lower bound is initial surface temperature, but upper bound is now different:

$$\mathrm{Y} = [T_{s,0}, T], \text{ where } T \text{ s.t. } h(T)(T - T_a) = 2\dot{q}_o \tag{15}$$

It is important to note, that in case of fluid gaining heat from surroundings (inlet fluid temperature is lower than ambient temperature), there are only two cases to consider, either (1) or (3), bacause $\Delta T_{sa,0}$ will be negative or equal to 0, and Y will be (13) or (15).